



An augmented Kalman filter for force identification in structural dynamics

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ABSTRACT

An augmented Kalman filter for force identification in structural dynamics is developed, in which the unknown forces are included in the state vector and estimated in conjunction with the states. Noise is modeled as a stochastic process and is assumed to be present not only on the measurements, but also on the state variables, thus accounting to some extent for modeling errors. This distinguishes the proposed technique from purely deterministic methods for force identification in which no errors are assumed on the states. To analyze the effect hereof on the quality of the identification, the results obtained with a commonly used recursive least-squares method, the Dynamic Programming algorithm, are compared to those obtained using the augmented filter in a laboratory experiment on an instrumented steel beam. It is shown how, in the collocated case, more accurate results can be obtained with the augmented filter due to its incorporation of modeling errors. In the non-collocated case, however, better solutions are produced by classical deterministic methods as Dynamic Programming in which only the forces are estimated, and not the states as well.

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1. Introduction

The general force identification problem deals with the reconstruction of unknown dynamic forces acting on a structure from its response in a limited number of sensors and a system model. The system model can be obtained by a priori dynamic system identification. The need to inversely determine excitation sources arises when direct measurements of the forces are not possible or very difficult. Examples include the determination of dynamic vehicle loads on bridges or of dynamic wind loads on tower structures. When reconstructing these forces using an inverse approach, uncertainty is introduced through the measured response (measurement errors) and the modeling assumptions (model errors).

In a structural dynamics context force identification problems were originally, in the 1980s, solved using the frequency response functions of the structure [1–5]. Since then many alternative methods have been developed, mostly formulated in the time domain and primarily following a deterministic approach. Surveys are given by Klinkov [6] and Nordström et al. [7]. In recent years, however, the focus has shifted toward the development of combined deterministic-stochastic methods. Most of these techniques have their origin in control engineering and differentiate themselves from purely deterministic techniques by modeling the noise as stochastic processes and assuming noise to be present not only on the measurements, but also on the state variables. Conversely, the difference with purely stochastic techniques is that the unknown force values are still considered to be deterministic quantities.

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One of the earliest applications of combined deterministic-stochastic techniques to force estimation problems in structural dynamics has been presented by Ma et al. [8]. Utilizing an algorithm originally developed by Tuan et al. [9] for inverse heat conduction problems, impulsive loads acting on mass–spring–damper chains were estimated using a Kalman filter in conjunction with a recursive least-squares estimator. An important limitation of the proposed technique is the assumption that the measurement data vector contains the displacements of all degrees of freedom of the structure, rendering the technique inapplicable in many practical situations. In [10] and [11] satisfactory results of force identification experiments on a physical plate and beam in the laboratory were presented, made possible by modeling these structures as single degree of freedom (SDOF) systems. The state matrices of the SDOF systems were constructed from a collocated and non-collocated measurement using the eigensystem realization algorithm (ERA). Tuan's algorithm was also employed by Loh et al. [12] for the identification of the forces exerted on a pile by the soil during a shake table test with dynamic soil–pile interaction. An improvement to the algorithm was suggested later [13], in which the fading factor used to negotiate between the adaptive capability of the least-squares estimator and a loss of estimation accuracy was adjusted at each time step according to the residual innovation sequence of the Kalman filter. Finally, in [14], the technique was extended to non-linear systems by using an extended Kalman filter in conjunction with the least-squares estimator.

Recently Hwang et al. [15] suggested a more practical method in which all unmeasured responses required for an accurate estimation of the external loads are estimated with a Kalman filter to subsequently determine the loads, using the complete state vector, by means of a pseudo-inverse. The equations are converted to modal space to identify modal loads, and a numerical application to wind load estimation is included. Another interesting online method from control engineering was applied to structural loads monitoring by Klinkov et al. [16]. In this method an augmented state vector containing the states and inputs is estimated using an observer technique in which an optimal observer is found by solving a linear matrix inequality (LMI). The method was successfully applied to estimate multiple forces applied by an impact hammer and wind fan to a two-story test rig in the laboratory.

In this contribution an augmented Kalman filter (AKF) technique is proposed for dynamic force identification in a combined deterministic-stochastic setting. It is shown how joint input-state estimation can be achieved by including the unknown forces in the state-vector and estimating this augmented vector using a standard Kalman filter. The effectiveness of the method is verified by means of laboratory experiments on an instrumented steel beam. In order to illustrate the effect of incorporating modeling errors in the solution of the inverse problem, the results obtained using the AKF are compared to those obtained using a traditional deterministic least-squares technique, namely that of Dynamic Programming (DP) [17,18].

2. Problem formulation

The proposed technique essentially consists of a standard Kalman filter [19] applied to an augmented state-space model in which the forces are added to the unknown state vector. The augmented model is developed first, followed by a presentation of the filter equations.

2.1. Augmented state-space model

The augmented state-space model is derived starting from the classical discrete-time state equation with unknown noise vector $\mathbf{w}_k$ as a component of the stochastic noise process $\{\mathbf{w}_k \in \mathbb{R}^{n_s}\}_{k=0}^{\infty}$, where n_s refers to the number of states and $\mathbf{A}$ and $\mathbf{B}$ denote the system matrices

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{p}_k + \mathbf{w}_k. \quad (1)$$

The state equation (1) is supplemented with an equation which directly relates the force vectors $\mathbf{p} \in \mathbb{R}^{n_p}$ at times k and $k+1$

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \boldsymbol{\eta}_k, \quad (2)$$

with $\boldsymbol{\eta}_k$ a component of the stochastic process $\{\boldsymbol{\eta}_k \in \mathbb{R}^{n_p}\}_{k=0}^{\infty}$. Eq. (2) allows the estimation of the force time history through an appropriate choice of the covariance matrix of the process $\boldsymbol{\eta}_k$. In fact, it will be shown how this covariance matrix can be used to regularize the problem and how its optimal choice can be made using standard regularization parameter estimation techniques, e.g. the L-curve.

By combining Eqs. (1) and (2) and redefining the state vector $\mathbf{x}^a$, where the superscript 'a' refers to augmented

$$\mathbf{x}_k^a = \begin{bmatrix} \mathbf{x}_k \\ \mathbf{p}_k \end{bmatrix}, \quad (3)$$

an augmented state equation is obtained

$$\mathbf{x}_{k+1}^a = \mathbf{A}_a \mathbf{x}_k^a + \boldsymbol{\zeta}_k. \quad (4)$$

The matrix $\mathbf{A}_a \in \mathbb{R}^{(n_s + n_p) \times (n_s + n_p)}$, is defined as

$$\mathbf{A}_a = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}, \quad (5)$$

and the noise vector $\zeta_k \in \mathbb{R}^{n_s + n_p}$, again assumed unknown, accounts for the modeling errors $\mathbf{w}_k \in \mathbb{R}^{n_s}$ as well as the force increment $\eta_k \in \mathbb{R}^{n_p}$ in Eq. (2)

$$\zeta_k = \begin{Bmatrix} \mathbf{w}_k \\ \eta_k \end{Bmatrix}. \quad (6)$$

The classical discrete time observation equation with unknown noise vector $\mathbf{v}_k$ becomes, in the augmented model

$$\mathbf{d}_k = \mathbf{G}_a \mathbf{x}_k^a + \mathbf{v}_k, \quad (7)$$

where $\mathbf{d} \in \mathbb{R}^{n_d}$ represents the measured data vector and the matrix $\mathbf{G}_a \in \mathbb{R}^{n_d \times (n_s + n_p)}$ is assembled from the output influence and direct transmission matrices $\mathbf{G}$ and $\mathbf{J}$ as follows:

$$\mathbf{G}_a = [\mathbf{G} \ \mathbf{J}]. \quad (8)$$

To summarize, the state and observation equation defining the augmented state-space model are formulated as follows:

$$\mathbf{x}_{k+1}^a = \mathbf{A}_a \mathbf{x}_k^a + \zeta_k, \quad (9)$$

$$\mathbf{d}_k = \mathbf{G}_a \mathbf{x}_k^a + \mathbf{v}_k. \quad (10)$$

The system matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{G}$ and $\mathbf{J}$ used in the identification can be obtained in different ways. Firstly, the modal parameters identified in an a priori modal test can be used to construct a data-driven state-space model [20]. The use of a data-driven model allows minimizing the modeling error, but is only possible when a sensor is present at each force location and when scaled modes can be identified using a driving-point measurement. Alternatively, the system matrices can be identified directly using system identification techniques, as has for instance been done by Liu et al. [10]. In this contribution a FE model, tuned using the modal parameters obtained by a priori modal analysis, will be used for the identification (see Section 3.1).

2.2. Kalman filter

The Kalman filter can be defined as a recursive linear state estimator designed to be optimal in a minimum-variance unbiased sense. Before presenting the filter expressions some definitions are given and the underlying assumptions clarified. Firstly, $\hat{\mathbf{x}}_{k|l}^a$ is defined as an estimate of $\mathbf{x}_k^a$ given $\{\mathbf{d}_n\}_{n=0}^l$. Note that according to this convention $\hat{\mathbf{x}}_{0|-1}^a$ refers to an initial estimate of $\mathbf{x}^a$ at time $k=0$. The error covariance matrix $\mathbf{P}_{k|l} \in \mathbb{R}^{(n_s + n_p) \times (n_s + n_p)}$ is defined as

$$\mathbf{P}_{k|l} = \mathbb{E}[(\mathbf{x}_k^a - \hat{\mathbf{x}}_{k|l}^a)(\mathbf{x}_k^a - \hat{\mathbf{x}}_{k|l}^a)^T], \quad (11)$$

and it is assumed that the initial error covariance matrix $\mathbf{P}_{0|-1}$ as well as the initial state estimate $\hat{\mathbf{x}}_{0|-1}^a$ are known. Notice that the initial state $\mathbf{x}_0^a$ is considered to be a random variable and does not have to be known exactly as is the case in most deterministic settings.

The noise processes $\{\mathbf{w}_k \in \mathbb{R}^{n_s}\}_{k=0}^\infty$, $\{\mathbf{v}_k \in \mathbb{R}^{n_d}\}_{k=0}^\infty$ and $\{\eta_k \in \mathbb{R}^{n_p}\}_{k=0}^\infty$ are assumed to be stationary, mutually uncorrelated stochastic processes with zero mean. Their covariances are assumed known and are represented by matrices $\mathbf{Q} \in \mathbb{R}^{n_s \times n_s}$, $\mathbf{R} \in \mathbb{R}^{n_d \times n_d}$ and $\mathbf{S} \in \mathbb{R}^{n_p \times n_p}$, respectively

$$\mathbb{E}\{\mathbf{w}_k \mathbf{w}_l^T\} = \mathbf{Q} \delta_{k-l}, \quad (12)$$

$$\mathbb{E}\{\mathbf{v}_k \mathbf{v}_l^T\} = \mathbf{R} \delta_{k-l}, \quad (13)$$

$$\mathbb{E}\{\eta_k \eta_l^T\} = \mathbf{S} \delta_{k-l}, \quad (14)$$

where δ_{k-l} is the Kronecker delta.

Since the so-called smoothness of the solution is regulated by the covariance matrix $\mathbf{S}$, this matrix can be viewed as a regularization matrix. The ill-conditioning of the problem can thus be treated by choosing an optimum regularization matrix $\mathbf{S}$. To obtain optimal values for the diagonal elements of $\mathbf{S}$ representing the variance on the force increments, standard regularization parameter estimation methods such as the L-curve [21] can be used. In the case of the augmented Kalman filter, however, the curve representing the smoothing norm (mostly taken as the square of the norm of the identified force) as a function of the error norm, does not exhibit its classical L-shape. This can be explained by elaborating these norms for the augmented filter. The smoothing norm remains the same as in the classical case where only the forces (without consideration of the states) are estimated, and can be written as $\sum_{k=1}^N \{\|\hat{\mathbf{p}}_k\|_2^2\}$. The error norm, however, now depends on the force as well as on the state estimates, and its value in the classical case, $\sum_{k=1}^N \{\|\mathbf{d}_k - (\mathbf{G}\mathbf{x}_k + \mathbf{J}\hat{\mathbf{p}}_k)\|_2^2\}$, is replaced by $\sum_{k=1}^N \{\|\mathbf{d}_k - (\mathbf{G}\hat{\mathbf{x}}_k + \mathbf{J}\hat{\mathbf{p}}_k)\|_2^2\}$ for the augmented Kalman filter. In situations where an optimal regularization matrix $\mathbf{S}$ exists, there is generally speaking very little variation in the error norm for different levels of regularization, since both the forces $\hat{\mathbf{p}}$ and the states $\hat{\mathbf{x}}$ can be adjusted to fit the measured data $\mathbf{d}$. The smoothing norm, on the other hand,

varies significantly with $\mathbf{S}$, and thus becomes the more important parameter. It was found that, if an optimal value for the diagonal entries of $\mathbf{S}$ exists, the smoothing norm is relatively insensitive to values of $\mathbf{S}$ within a given optimal range. This behavior will be illustrated in the experimental examples. For simplicity the term L-curve, although lacking its classical L-shape, is maintained in the remainder of the text.

The choice of the diagonal elements of the covariance matrices $\mathbf{Q}$ and $\mathbf{R}$ should be based on the order of magnitude of the state vector and the accuracy of the sensors, respectively. Generally a constant value is used on these diagonals. The solution is, however, much less sensitive to these matrices than to the regularization matrix $\mathbf{S}$. The solution's sensitivity to the covariance matrix of the initial state error $\mathbf{P}_{0|-1}$ depends on the size of this error, which is unknown, but in most cases it suffices to assign very large values to the diagonal of this matrix which has the effect of treating the initial errors as large and causing the filter to ignore the initial estimations.

The Kalman filter equations for the discrete-time state-space system of Eqs. (9) and (10) are now presented in terms of the measurement update, where the previously predicted state estimate is updated with the new observation, and the time update, where the state is advanced based on the model equations:

Measurement update:

$$\mathbf{L}_k = \mathbf{P}_{k|k-1} \mathbf{G}_a^T (\mathbf{G}_a \mathbf{P}_{k|k-1} \mathbf{G}_a^T + \mathbf{R})^{-1}, \quad (15)$$

$$\hat{\mathbf{x}}_{k|k}^a = \hat{\mathbf{x}}_{k|k-1}^a + \mathbf{L}_k (\mathbf{d}_k - \mathbf{G}_a \hat{\mathbf{x}}_{k|k-1}^a), \quad (16)$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{L}_k \mathbf{G}_a \mathbf{P}_{k|k-1}, \quad (17)$$

Time update:

$$\hat{\mathbf{x}}_{k+1|k}^a = \mathbf{A}_a \hat{\mathbf{x}}_{k|k}^a, \quad (18)$$

$$\mathbf{P}_{k+1|k} = \mathbf{A}_a \mathbf{P}_{k|k} \mathbf{A}_a^T + \mathbf{Q}_a. \quad (19)$$

Note that in conjunction with the augmented noise vector ζ_k of Eq. (9), the following augmented covariance matrix $\mathbf{Q}_a \in \mathbb{R}^{(n_s + n_p) \times (n_s + n_p)}$ is used:

$$\mathbf{Q}_a = \begin{bmatrix} \mathbf{Q} & \mathbf{0} \\ \mathbf{0} & \mathbf{S} \end{bmatrix}. \quad (20)$$

2.3. Dynamic programming (DP)

Since noise is assumed on the state variables in a combined deterministic-stochastic approach, modeling errors (in the system matrices $\mathbf{A}$ and $\mathbf{B}$) as well as forces applied at locations other than the input locations considered in the identification (errors in the input matrix $\mathbf{B}\mathbf{p}_k$) are to some extent accounted for. This distinguishes the present technique from purely deterministic methods in which the modeling error $\mathbf{w}_k$ in Eq. (1) is assumed to be zero. To analyze the effect hereof on the quality of the identification the DP algorithm, a commonly used recursive least-squares method, is used to identify forces in a laboratory experiment and the results are compared to those obtained with the AKF.

The DP algorithm was originally proposed by Trujillo [22,18] and has been successfully used for the reconstruction of forces from vibration measurements in the fields of structural dynamics – for loads at fixed positions [23,24,22] as well as moving loads [25,31] – and heat conduction [22]. It was later extended by Nordström [17] to enable higher order input estimations and the use of an output relation which includes direct transmission. The algorithm essentially consists of minimizing the following least-squares cost function in the presence of a Tikhonov penalty term:

$$E = \|\mathbf{d}_k - \mathbf{G}\mathbf{x}_k - \mathbf{J}\mathbf{p}_k\|_2^2 + \lambda \|\mathbf{p}_k\|_2^2, \quad (21)$$

where the minimization is done iteratively, starting from the last time step.

In this contribution the results obtained with Nordström's extended algorithm, implemented as in [26], are compared with those obtained using the AKF.

3. Experimental results

To verify the effectiveness of the AKF a series of laboratory experiments are performed on an instrumented steel I-beam. The beam has an IPE100 cross section with a height of 100 mm, a width of 55 mm, and web and flange thicknesses of 4.1 mm and 5.7 mm, respectively. It has a length of 3 m, and steel plates of 150 mm × 150 mm × 15 mm welded to its ends. At both ends the beam is suspended from a steel frame using flexible springs to simulate free-free boundary conditions. A series of accelerometers are placed along the beam to record its response to different forces, applied with an instrumented hammer and hydraulic shaker. The proposed method as well as the DP algorithm are used to reconstruct the force vector from the measured data, with the positions of the forces assumed to be known.

3.1. Model description

The system matrices $\mathbf{A}$ and $\mathbf{B}$ are constructed from the FE model matrices by departing from the continuous-time governing equations of motion for a linear system discretized in space

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{f}(t) = \mathbf{S}_p(t)\mathbf{p}(t), \quad (22)$$

where $\mathbf{u} \in \mathbb{R}^{n_{\text{DOF}}}$ is the vector of degrees of freedom, $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ denote the mass, damping and stiffness matrix, respectively, and $\mathbf{p}(t)$ is the excitation. The excitation is factorized into a selection matrix $\mathbf{S}_p(t) \in \mathbb{R}^{n_{\text{DOF}} \times n_p}$ specifying the locations of the forces, and the vector $\mathbf{p}(t) \in \mathbb{R}^{n_p}$, representing the n_p force time histories. These governing equations are discretized in time into their previously presented state-space form, repeated here for convenience

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{p}_k. \quad (23)$$

The specific forms of $\mathbf{A}$ and $\mathbf{B}$ and the content of the state vector depend on the time discretization scheme. Here a zero-order hold is assumed on the force $\mathbf{p}(t)$ and the discrete-time state-space model is obtained by defining the original, i.e. not the augmented, state-vector as

$$\mathbf{x}_k = \begin{bmatrix} \mathbf{u}_k \\ \dot{\mathbf{u}}_k \end{bmatrix} \quad (24)$$

and using the exponential matrix

$$\mathbf{A} = e^{\mathbf{A}_c \Delta t}, \quad \mathbf{B} = [\mathbf{A} - \mathbf{I}]\mathbf{A}_c^{-1}\mathbf{B}_c, \quad (25)$$

where

$$\mathbf{A}_c = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{S}_p \end{bmatrix} \quad (26)$$

and the subscript 'c' denotes continuous time. To arrive at the observation Eq. (7), the following general relation between the data vector and the measurement quantities is considered:

$$\mathbf{d}(t) \approx \mathbf{S}_a\ddot{\mathbf{u}}(t) + \mathbf{S}_v\dot{\mathbf{u}}(t) + \mathbf{S}_d\mathbf{u}(t), \quad (27)$$

where $\mathbf{S}_a$, $\mathbf{S}_v$ and $\mathbf{S}_d \in \mathbb{R}^{n_d \times n_{\text{DOF}}}$ are selection matrices for acceleration, velocity and displacement, respectively. When acceleration, velocity or displacement is measured, they thus take the form of boolean vectors in which the measurement locations are specified. If strains are measured, they specify difference relations, in which the strains are formulated as a linear combination of the displacement states. Using Eq. (22) and the definition (24) of the state vector, Eq. (27) is transformed into its state-space form:

$$\mathbf{d}_k \approx \mathbf{G}\mathbf{x}_k + \mathbf{J}\mathbf{p}_k, \quad (28)$$

where the output influence and direct transmission matrices $\mathbf{G}$ and $\mathbf{J}$ then relate to the FE model matrices as follows:

$$\mathbf{G} = [\mathbf{S}_d - \mathbf{S}_a\mathbf{M}^{-1}\mathbf{K} \quad \mathbf{S}_v - \mathbf{S}_a\mathbf{M}^{-1}\mathbf{C}], \quad \mathbf{J} = \mathbf{S}_a\mathbf{M}^{-1}\mathbf{S}_p. \quad (29)$$

Eq. (29) implies that the direct transmission matrix $\mathbf{J}$ is non-zero when the measured data vector $\mathbf{d}$ contains at least one acceleration time history. Conversely, when only displacement or strain is measured, no direct transmission exists.

To reduce the model order and, therefore, the computational effort, modally reduced mass, stiffness and damping matrices are used when constructing the state matrices in Eqs. (25) and (29). The modally reduced mass and stiffness matrices are constructed using the eigenmodes and frequencies obtained with the (updated) FE model, whilst the experimental damping ratios are used to construct the modal damping matrix.

3.2. Measurement setup

The vibration response of the beam was measured using 19 accelerometers (PCB 338B35, sensitivity ± 100 mV/g). Each test was performed in two different setups, where 15 accelerometers were moved from one setup to the other so that in total 34 accelerations at different positions of the beam were measured. Only vertical accelerations on the top flange were measured, at both transversal ends of the beam at 17 different cross sections (Figs. 1 and 2). The forces were applied eccentrically with an impact hammer (PCB 086C03, mass 0.136 kg) and a shaker (MB Modal 50A, mass 22.68 kg) at one of the free-free beam ends. In order to measure the force applied by the shaker on the beam, a load cell was placed in between the beam and the stinger. All sensors were connected to a NI PXI-1050 front-end, which in turn was controlled by a portable computer. The sampling frequency was set to 10 kHz.

3.3. Modal analysis

In order to identify the modal parameters, multiple hammer impacts were applied to the beam. A concatenation of 40 s of measured data was processed to extract the rigid body and elastic modes of the steel beam. For the rigid body modes the

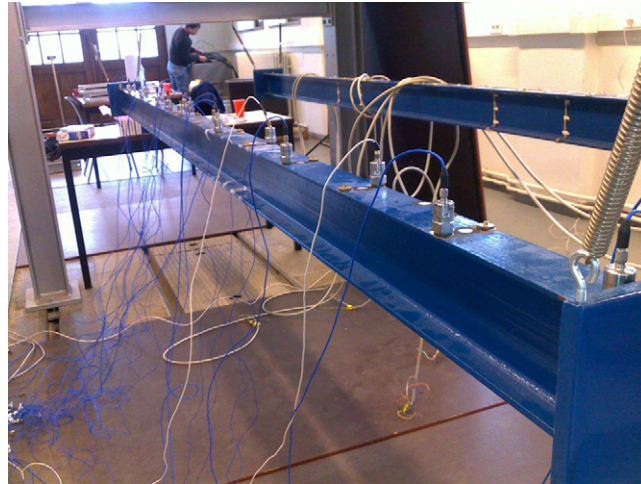


Fig. 1. The measurement setup.

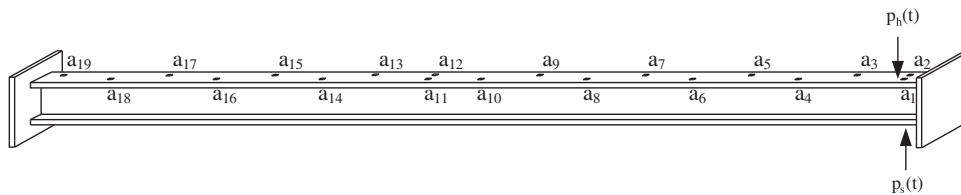


Fig. 2. Positions of the sensors and applied forces.

Table 1

Undamped eigenfrequencies (f_{ud}), damping ratios (ξ), modal phase collinearities (MPC), mean phase values (MP) and deflection type of the identified modes of the steel beam in the frequency range 0–900 Hz.

Mode no.	f_{ud} (Hz)	ξ (%)	MPC (–)	MP (°)	Type
1	1.77	0.36	0.955	4.0	Rigid body
2	2.70	0.19	0.956	3.0	Rigid body
3	4.92	0.40	0.994	5.0	Rigid body
4	30.81	0.12	1.000	0.5	1st torsion
5	60.90	0.05	1.000	3.0	1st bending
6	71.20	0.32	1.000	2.9	2nd torsion
7	128.13	0.15	1.000	8.0	3rd torsion
8	173.86	0.04	1.000	0.4	2nd bending
9	200.49	0.12	1.000	1.0	4th torsion
10	293.51	0.13	0.996	4.3	5th torsion
11	340.95	0.02	0.999	0.6	3rd bending
12	402.56	0.11	0.998	1.7	6th torsion
13	535.38	0.12	0.991	2.5	7th torsion
14	556.70	0.03	0.998	1.3	4th bending
15	805.65	0.04	0.995	1.8	5th bending

signals were low-pass filtered with an eighth-order Chebychev Type I filter with a cutoff frequency of 40 Hz in both the forward and reverse direction to remove all phase distortion. Subsequently the data was re-sampled at 100 Hz. The elastic modes were obtained by low-pass filtering the signals with the same filter and a cutoff frequency of 2000 Hz in both the forward and reverse direction. The sampling rate was reduced to 5000 Hz.

The two different setups have been processed independently using the reference-based data-driven combined deterministic-stochastic subspace identification (CSI-data/ref.) algorithm [27]. The following parameter choices were made: half the number of block rows in the data Hankel matrix $\iota = 100$; the driving point acceleration and a second acceleration common to both setups were chosen as reference outputs; a model order range from 4 to 140 in steps of 4 was chosen for the construction of the stabilization diagrams.

Table 1 lists the identified modes in the frequency range 0–900 Hz. All modes are very lightly damped with damping ratios below 0.5% of critical. The modal phase collinearity (MPC) values give an indication of the real character of a mode shape [28]. When the MPC is close to one, as for all modes listed here, the identified modes are nearly perfect real normal

modes. The mean phases (MP) of the mass-normalized mode shapes are also given. For perfect real normal mass-normalized mode shapes the mean phases should be zero. From the MPC and MP values it can therefore be concluded that the modal parameters presented in Table 1 are of high quality. The natural frequencies in the table, in the range 0–900 Hz, are subsequently used to tune the Young's modulus of the steel E and the stiffness of the springs k in the FE model. A comparison between the identified eigenfrequencies and those from the FE model with E and k set to 207 GPa and 1350 N/m, respectively, are shown in Table 2. Except for the torsional modes, whose contribution to the acceleration response considered is almost negligible, a good correspondence is found. The 15 modes from the FE model – 3 rigid body, 5 bending and 7 torsion – are used to construct the reduced model for the force identification. In order to include only frequencies within the range of the identified modes, all the data used in the inverse calculations are resampled at a lower rate. Using a decimation factor of 5, the data are low-pass filtered using a Chebychev Type I filter at 800 Hz and subsequently resampled at 2 kHz.

To assess the accuracy of the model one of the measured hammer impacts is used as input in a forward analysis and the calculated response is compared with the measured acceleration. Figs. 3 and 4 show such a comparison for the driving point acceleration in the time and frequency domain, respectively. The dominant frequency components in the data, i.e. the five largest peaks in Fig. 4a corresponding to the first five bending modes, are well predicted by the model. Note that it is also for these bending mode frequencies that the best correspondence between the FE model and the experimental modal analysis results is found (cf. Tables 1 and 2).

Table 2

Comparison of the eigenfrequencies identified in the experimental modal analysis and those from the FE model.

Eigenfrequency [Hz]	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
Identified	1.77	2.70	4.92	30.81	60.90	71.20	128.13	173.86
FE	1.78	2.79	3.11	31.56	61.51	74.94	136.02	174.57
	f_9	f_{10}	f_{11}	f_{12}	f_{13}	f_{14}	f_{15}	
Identified	200.49	293.51	340.95	402.56	535.38	556.70	805.65	
FE	216.91	319.07	341.88	443.27	589.82	557.37	807.07	

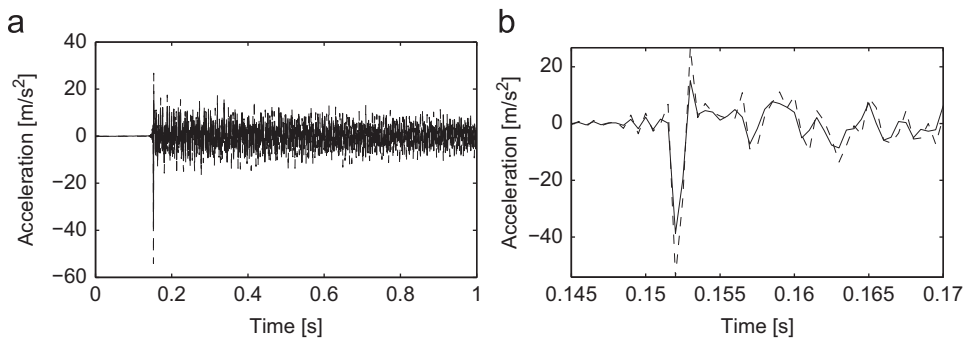


Fig. 3. (a) Complete and (b) detail of the measured (–) and calculated (– –) time history of the driving point acceleration due to a hammer impact.

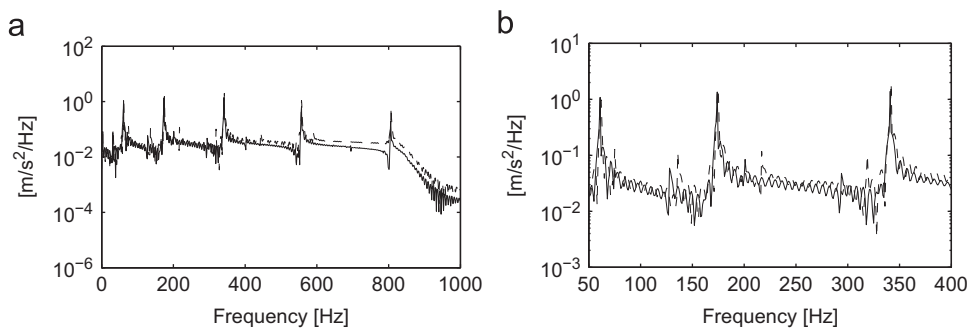


Fig. 4. (a) Complete and (b) detail of the measured (–) and calculated (– –) frequency spectrum of the driving point acceleration due to a hammer impact.

3.4. Force identification

In the following the AKF is used to identify a hammer impact and a sine-sweep applied with a hydraulic shaker, and the results are compared to those obtained using DP. In accordance with what they represent, the order of magnitude of the square roots of the diagonal elements of $\mathbf{R}$ and $\mathbf{Q}$ is chosen to correspond to a small percentage of the highest peaks in the measured response and the states (displacements/velocities), respectively. An optimal order of magnitude for the covariance matrix of the force increment $\mathbf{S}$, Eq. (14), is obtained using the L-curve method [21].

In Fig. 2 each sensor is assigned a label to facilitate the indication of which sensors are used to identify the forces. The figure also shows the positions of the hammer impact $p_h(t)$ and the shaker force $p_s(t)$.

3.4.1. Identification of a hammer impact

In the first analysis an attempt is made to reconstruct the hammer impact using a single sensor located in the immediate vicinity of the impact location, i.e. the response measured in sensor a_1 , which was presented in Figs. 3 and 4. The covariance matrices $\mathbf{P}_{0|-1}$, $\mathbf{Q}$ and $\mathbf{R}$ from Eqs. (11)–(13) are assigned values of 10^{-6} , 10^{-6} and 10^{-1} on the diagonal, respectively. Since acceleration is measured, the direct transmission term in Eq. (28) which represents a direct influence of the forces $\mathbf{p}_k$ on the data $\mathbf{d}_k$ at a time step k , is non-zero. Furthermore, since measuring the acceleration a_1 for the hammer impact can be considered a collocated measurement, the input has instant and distinguishable influence on the output. When only the forces are estimated as is the case with DP, such problems fulfill the condition for the existence of a unique solution and are, in fact, well-posed [29]. The solution is said to be stable, i.e. a continuous function of the data, and the noise on the data is seen back in the solution without any amplification, as would have been the case for an ill-posed problem. In accordance herewith, the Tikhonov regularization parameter is set to zero for the force estimation using DP. It is mentioned that when strain or displacement is measured, there is no direct transmission and well-posedness can never be achieved because the forces are only indirectly related to the data via their effect on some linear combination of the states (cf. Eq. (28)).

Establishing ill-/wellposedness for the augmented Kalman filter is more complicated. The existence of a unique solution for the augmented state vector is determined by the observability property of the matrix pair $\{\mathbf{A}_a, \mathbf{G}_a\}$. The rank of the observability matrix indicates whether the augmented state vector can be determined uniquely in finite time. For all problems treated here the singular values of the observability matrix were found to decay gradually to zero, so that it is difficult to establish a notion of rank and, therewith, well-posedness for these problems. For the current analysis with one collocated sensor the L-curve converges to a single point as the regularization parameter is increased. This implies no regularization since, as opposed to DP, a smaller regularization parameter would favor a smoother solution and vice versa (cf. η_k in Eq. (2)). In what follows an arbitrarily large value of 10^{30} is assigned to the diagonal entries of the covariance matrix $\mathbf{S}$ of Eq. (14).

The measured and identified forces using the AKF and DP are compared in Figs. 5 and 6 in the time and frequency domain, respectively. A detail of the time history at the instant of the impact is presented in Fig. 5b. Except for a slight drift after the impact in the force identified using the AKF, the quality of the reconstruction is seen to be almost identical for the two algorithms. The drift can be attributed to the use of acceleration data in the identification. Since acceleration is not sensitive to any quasi-static component in the input, solutions containing such components satisfy the data equally well as the sought solution, in which these components are zero. The phenomenon is not found in the force identified using DP, since the Tikhonov penalty term $\lambda \|\mathbf{p}_k\|$ in the DP cost function, Eq. (21), penalizes solutions having a large smoothing norm. The regularization with the Kalman filter, being in fact an online method in which the force at each time step is identified based on a measurement and time update, does not ‘globally’ minimize the smoothing norm over the analyzed period, but rather minimizes the variation in the force from one time step to the next, thereby allowing the observed drift.

Both reconstructed forces exhibit some residual ringing after the impact, a phenomenon that has also been observed by other researchers [30]. The ringing is present in the frequency spectrum, Fig. 6, as spurious peaks at the eigenfrequencies

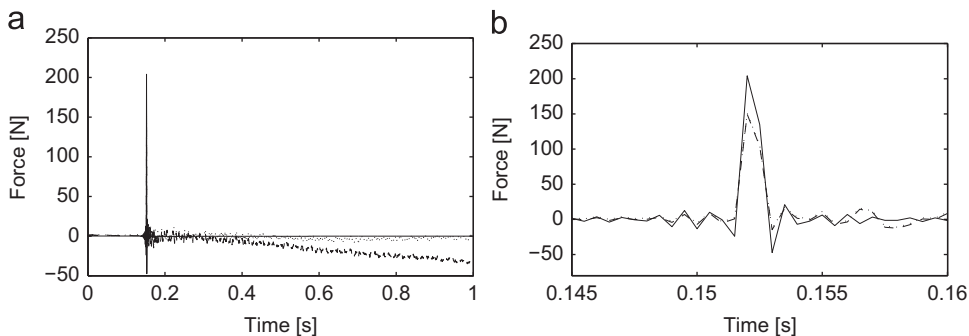


Fig. 5. (a) Complete and (b) detail of the time history of the measured (–) hammer impact versus the reconstructed force obtained with a single collocated measurement using the AKF (– –) and DP (···) and no regularization.

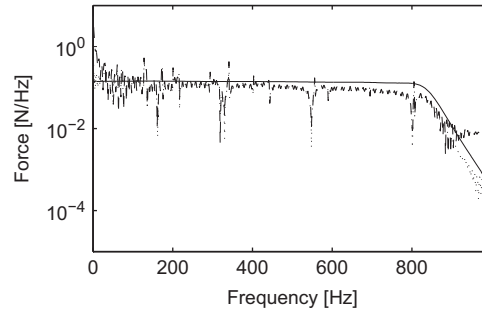


Fig. 6. Frequency spectrum of the measured (–) hammer impact versus the reconstructed force obtained with a single collocated measurement using the AKF (– –) and DP (···) and no regularization.

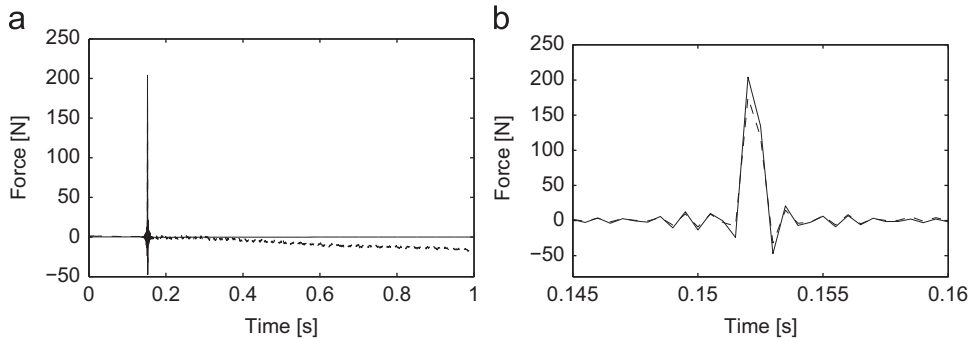


Fig. 7. (a) Complete and (b) detail of the measured (–) versus reconstructed (– –) time history of the hammer impact obtained with the AKF using six sensors and an optimal amount of regularization.

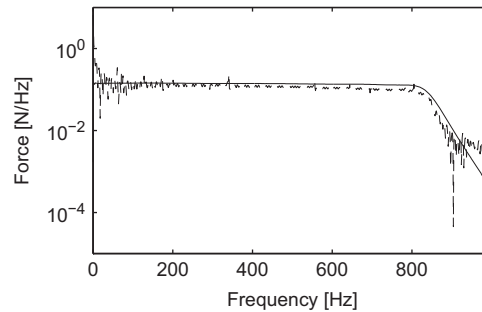


Fig. 8. Frequency spectrum of the measured (–) versus reconstructed (– –) hammer impact obtained with the AKF using six sensors and an optimal amount of regularization.

of the beam and can be ascribed to increased ill-conditioning at resonance. It was found that, using the AKF, these peaks can be eliminated by increasing the number of sensors. In Figs. 7 and 8 the results obtained using an additional five accelerometers, a_3 to a_7 (cf. Fig. 2) are presented. The reconstruction is significantly improved with peaks remaining only at the lowest eigenfrequencies in Fig. 8 and with the impact itself identified almost perfectly in Fig. 7b.

The same inverse problem, using data from six sensors, is now solved using the DP algorithm and the results are presented in Figs. 9 and 10. The identified force is comparable to the one obtained using a single sensor in Figs. 5 and 6 and the solution is not improved as in the case of the AKF. The accuracy of the forces identified with a deterministic method like DP is limited by the accuracy with which the model can predict the measured data. Since these methods do not have the ability to account for modeling errors and to adjust the states and forces based on the measured data, a discrepancy between the measured and predicted data always remains. The AKF, however, manages to accurately predict the data, i.e. the error between $\mathbf{G}_a \hat{\mathbf{x}}^a$ and the measured data vector $\mathbf{d}$ is found to be negligible.

The square of the norm of this error over the analyzed period, $\sum_{k=1}^N \{\|\mathbf{d}_k - \mathbf{G}_a \hat{\mathbf{x}}_k^a\|_2^2\}$, for different levels of regularization, can be read on the abscissa of the L-curve. Fig. 11 shows this curve for the identification with the AKF using six sensors. Some of the regularization levels, i.e. the value assigned to the diagonal entries of $\mathbf{S}$, are indicated on the curve. Whereas for the identification using a single collocated measurement the L-curve would converge to a single point as this value was

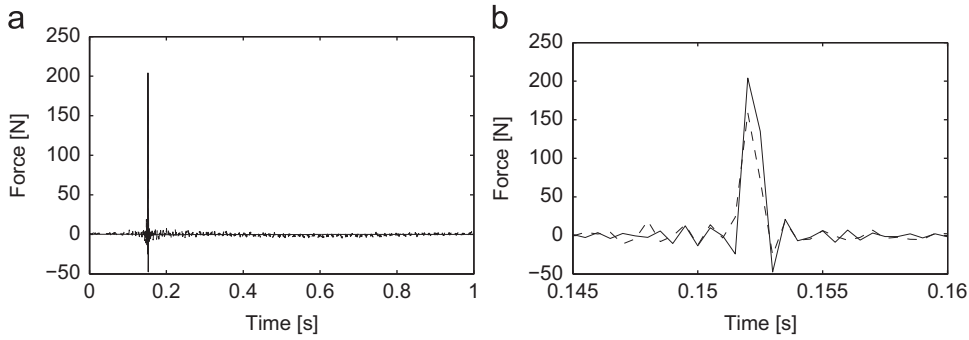


Fig. 9. (a) Complete and (b) detail of the measured (–) versus reconstructed (– –) time history of the hammer impact obtained with DP using six sensors and no regularization.

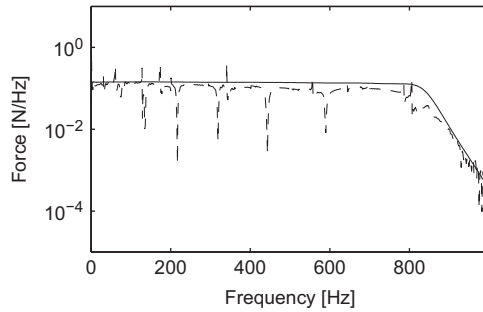


Fig. 10. Frequency spectrum of the measured (–) versus reconstructed (– –) hammer impact obtained with DP using six sensors and no regularization.

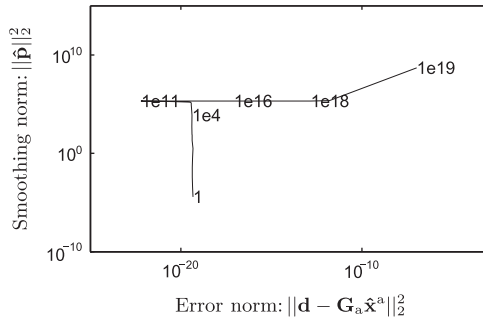


Fig. 11. The L-curve for the identification with the AKF using six sensors.

increased, the curve in Fig. 11 indicates that regularization becomes necessary when more sensors are used. It also illustrates the fact that, compared to the smoothing norm, the error norm is relatively insensitive to the regularization parameter as explained earlier. Here the smoothing norm and, in fact, the solution, are seen to stabilize for regularization parameters above 10^6 with relatively very little change in the error norm for higher values of the regularization parameter.

In the deterministic case, however, well-posedness still holds and the L-curve for the DP algorithm in the case where six sensors are used is again found to converge to a single point as the regularization parameter is decreased. The DP results of Figs. 9 and 10 are then also obtained with the regularization parameter set to zero.

In many cases the force location may not be accessible for measurement and one would be forced to revert to non-collocated measurements. In a last analysis with the hammer impact the effect hereof on the quality of the identification will be illustrated. In the non-collocated case, the existence of a unique solution (and thus well-posedness) will in the case of DP depends on the rank of the direct transmission matrix $\mathbf{J} \in \mathbb{R}^{n_d \times n_p}$. For the current example, where $n_p = 1$ and data from six non-collocated accelerometers a_5 to a_{10} will be used to identify the force, well-posedness still holds and no regularization has to be applied. Results are plotted in Figs. 12 and 13 (dotted line). Due to an increased time delay between the input and the measured response in some of the sensors these results are of a slightly lesser quality than those of Figs. 9 and 10, but the impact is still reasonably well identified.

When the same six non-collocated measurements are used to reconstruct the force by means of the AKF, the results are much worse (cf. dashed lines in Figs. 12 and 13). The corresponding L-curve is shown in Fig. 14, for the same range of

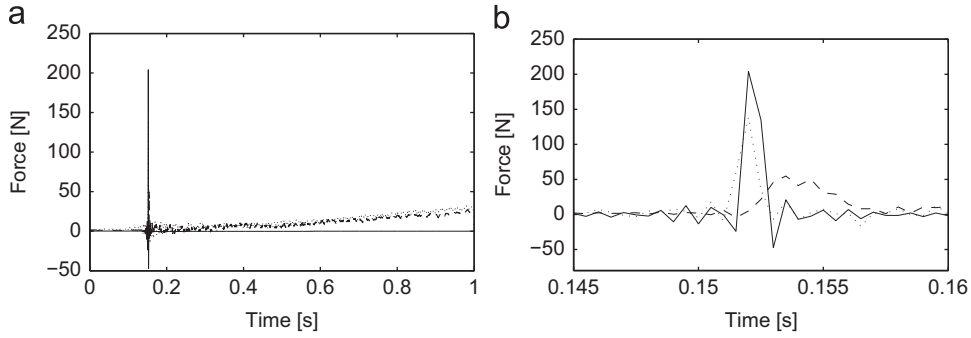


Fig. 12. (a) Complete and (b) detail of the time history of the measured (–) hammer impact versus the reconstructed force obtained with six non-collocated sensors using the AKF (– –) and DP (···).

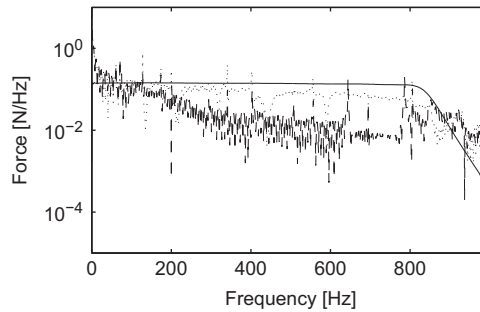


Fig. 13. Frequency spectrum of the measured (–) hammer impact versus the reconstructed force obtained with six non-collocated sensors using the AKF (– –) and DP (···).

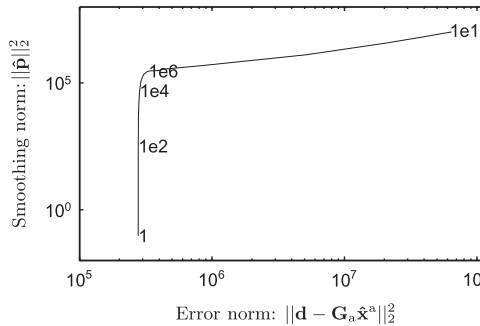


Fig. 14. The L-curve for the identification with the AKF using six non-collocated sensors.

regularization parameters as in the collocated case (Fig. 11). The curve clearly does not show the convergence to a stable smoothing norm with decreasing error norm observed in the collocated case, and no optimal regularization parameter exists. The order of magnitude of the error norm has also increased significantly. The reason for these poor results can be attributed to the ill-conditioning which, as explained earlier, is inherent to the AKF formulation in which both states and forces are estimated. When the distance between the force location and the sensors increases, the ill-conditioning increases, a statement confirmed by observing that the condition number of the observability matrix has, in the non-collocated case, increased by a factor of 6 compared to the previously analyzed collocated case with six sensors.

3.4.2. Identification of a sine-sweep

In this second example a sine-sweep is applied to the beam using the shaker. The maximum amplitude of the measured force is approximately one order of magnitude smaller than for the hammer impact considered previously. The covariance matrices $\mathbf{P}_{0|-1}$, $\mathbf{Q}$ and $\mathbf{R}$ are thus adjusted to have, respectively, values of 10^{-7} , 10^{-7} and 10^{-2} on the diagonal. The measured applied force is shown in Figs. 15 and 16 together with the forces identified with a single sensor, a_1 , using the AKF and DP, respectively. No regularization is applied in both cases. The reconstructed forces are detrended to remove any low-frequency drift.

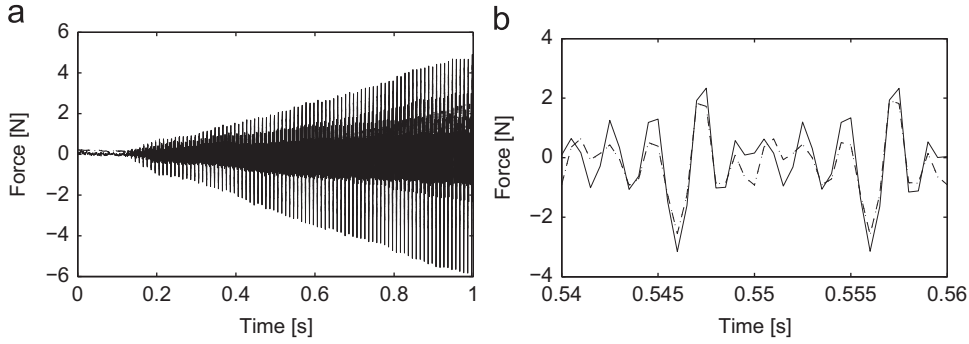


Fig. 15. (a) Complete and (b) detail of the time history of the measured (–) force versus the reconstruction obtained with a single collocated measurement using the AKF (– –) and DP (···) and no regularization.

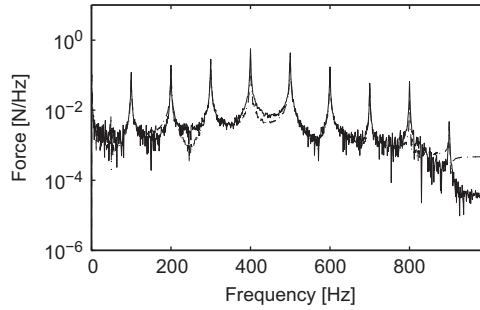


Fig. 16. Frequency spectrum of the measured (–) force versus the reconstruction obtained with a single collocated measurement using the AKF (– –) and DP (···) and no regularization.

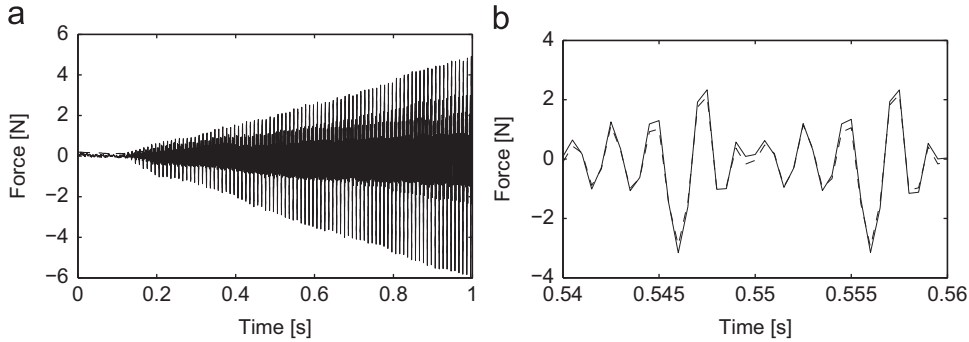


Fig. 17. (a) Complete and (b) detail of the measured (–) versus reconstructed (– –) time history of the force obtained with the AKF using six sensors and an optimal amount of regularization.

Since only discrete frequencies are excited, the increased ill-conditioning at resonance has a smaller effect than in the case of the hammer impact. The quality of the reconstruction is good, even with no overdetermination, and is again seen to be almost identical for the AKF and DP.

As in the previous example the accuracy of the identification using the AKF only surpasses that of DP when, in a collocated experiment, the number of sensors is increased. In Figs. 17 and 18 the detrended reconstruction obtained with the AKF using the same six sensors as in the previous example and an optimal amount of regularization is presented in the time and frequency domain, respectively. In Figs. 19 and 20 results are presented for the same identification problem solved using DP and no regularization. The conclusions drawn for the hammer impact in the previous section remain valid.

Finally it is mentioned that the six sensors used for the reconstruction of the impact and sine-sweep were chosen so as to achieve a minimum time delay between the input and the measured response in the sensor. For the sine-sweep this lead to a L2 error norm between the measured and identified forces, $\sqrt{\sum_{k=1}^N (\hat{p}_i - p_i)^2}$, of 9.27 and 25.69 for the AKF and DP, respectively. The results for a series of alternative groupings of six sensors are presented in Table 3. Except for the last three combinations in the table, all groupings include the collocated measurement a_1 . From the table it can first be concluded that the results obtained with the chosen six sensors are optimal. Secondly, in accordance with the observation

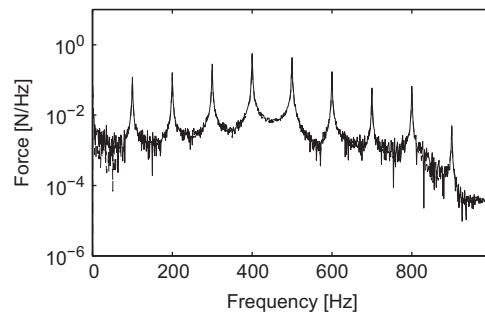


Fig. 18. Frequency spectrum of the measured (–) versus reconstructed (– –) force obtained with the AKF using six sensors and an optimal amount of regularization.

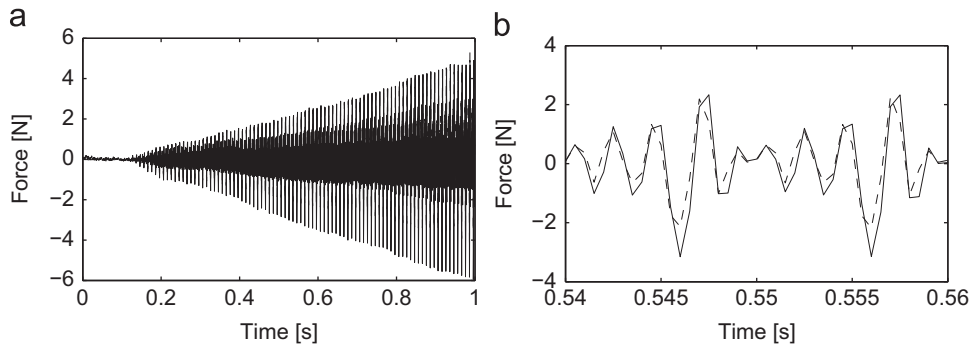


Fig. 19. (a) Complete and (b) detail of the measured (–) versus reconstructed (– –) time history of the force obtained with DP using six sensors and no regularization.

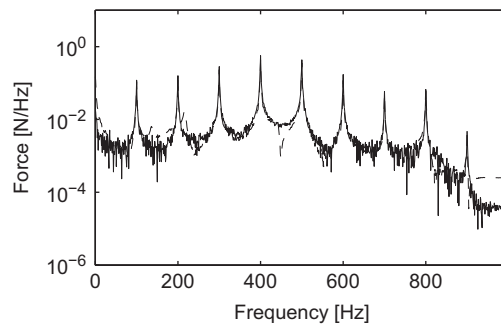


Fig. 20. Frequency spectrum of the measured (–) versus reconstructed (– –) force obtained with DP using six sensors and no regularization.

Table 3

Norm of the error between the measured and reconstructed forces obtained with different groups of sensors.

Sensor labels	AKF	DP	Sensor labels	AKF	DP
$a_1, a_3, a_4, a_5, a_6, a_7$	9.27	25.69	$a_1, a_3, a_6, a_9, a_{13}, a_{16}$	19.79	28.49
$a_1, a_4, a_5, a_6, a_7, a_8$	11.16	23.86	$a_1, a_5, a_8, a_{13}, a_{16}, a_{19}$	21.12	27.34
$a_1, a_5, a_6, a_7, a_8, a_9$	25.08	25.54	$a_1, a_5, a_8, a_{11}, a_{15}, a_{18}$	27.40	21.18
$a_1, a_6, a_7, a_8, a_9, a_{10}$	20.01	24.15	$a_1, a_4, a_7, a_{13}, a_{16}, a_{19}$	12.44	33.11
$a_1, a_9, a_{10}, a_{11}, a_{13}, a_{14}$	22.45	28.48	$a_1, a_4, a_7, a_{11}, a_{15}, a_{18}$	10.98	27.15
$a_1, a_{10}, a_{11}, a_{13}, a_{14}, a_{15}$	28.28	29.66	$a_3, a_4, a_5, a_6, a_7, a_8$	68.65	37.49
$a_1, a_{11}, a_{13}, a_{14}, a_{15}, a_{16}$	12.00	24.07	$a_4, a_5, a_6, a_7, a_8, a_9$	450.3	38.28
$a_1, a_3, a_5, a_8, a_{13}, a_{16}$	24.56	31.90	$a_5, a_6, a_7, a_8, a_9, a_{10}$	497.4	52.53

made earlier in the impact analysis, the AKF is seen to perform very poorly when no collocated measurement is available. The quality of the identification decreases much faster for the AKF than for the DP algorithm with an increase in the distance between the force and the sensors.

4. Conclusions

In this paper an augmented Kalman filter was presented for dynamic force identification in a combined deterministic-stochastic setting. The results obtained with the AKF in a laboratory experiment on a steel beam were compared to those obtained using Dynamic Programming, a traditional deterministic LS method. It was concluded that when a collocated measurement can be performed, the AKF is able to outperform the DP algorithm. Due to its ability to incorporate modeling errors into the identification process, the results obtained with the augmented filter can be considerably improved by increasing the number of sensors used to reconstruct the unknown force. This contrasts significantly with deterministic methods like DP in which the accuracy of the solution is bounded by the accuracy of the model.

When, however, measurements can only be performed at locations relatively far removed from the force location, the AKF fails to predict the applied force, whereas reasonable results can still be obtained with DP. The AKF's failure in these cases can be attributed to the fact that not only the forces, but also the complete state vector is estimated, resulting in a rapidly increasing level of ill-conditioning as the sensors are moved away from the force location.

Another drawback of the AKF concerns the identification of spurious low-frequency components when acceleration is measured. These drifts are caused by acceleration's insensitivity to any quasi-static component in the input. Whereas any identification technique in which Tikhonov regularization is applied automatically penalizes low-frequency components by means of a smoothing norm calculated over the complete time window, regularization with the Kalman filter implies only a local minimization of this norm at each time step. The minimization of this drift by incorporating for instance strain measurements in the data is a topic of ongoing research.

Although no numerical instabilities were encountered in this study, such problems have been reported in the past for both the Kalman filter and DP algorithms. Given the vast experience in state estimation using Kalman filter techniques, however, tried and tested methods for the stabilization of the Kalman filter exist and can be directly applied to the AKF. To the author's knowledge no stabilization techniques have up to now been presented for the extended DP algorithm.

Finally attention is drawn to the fact that, although the Kalman filter itself presents an on-line method for state estimation, the AKF can only be applied off-line since an optimal regularization parameter for the period under consideration has to be calculated.

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